

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building production agentic LLM systems and RAG pipelines, with a POC-builder mindset and strong track record of turning AI concepts into measurable production outcomes. Eager to bring that depth to Boomi's enterprise automation layer: translating customer integration requirements into AI-powered proof-of-concepts, demos, and onboarding experiences. Proven metrics: ~35% retrieval precision improvement, >30% hallucination reduction, ~30% fewer post-deployment defects. Python + LangChain + Docker proficient; remote-ready.

CORE COMPETENCIES

Agentic LLM Systems (LangChain, CrewAI-pattern) • RAG Pipeline Design • POC & Demo Development • Customer Technical Onboarding • Python & FastAPI • REST API Integration • Docker & CI/CD • Prompt Engineering • Observability & Logging • Enterprise AI Workflows

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting | USA

- Built production **agentic LLM workflows** for enterprise healthcare use cases (multi-step queries, eligibility automation, documentation clarification) using LangChain; improved agent response stability ~25% and reduced hallucinations >30%.
- Designed and deployed **RAG systems** with recursive semantic chunking, transformer embeddings, and evaluation pipelines; improved contextual retrieval precision ~35%, reduced irrelevant retrieval >30%.
- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with Docker; reduced post-deployment defects ~30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for customer risk scoring; improved recall 15-20% on high-risk categories.
- Built preprocessing pipelines (Pandas, NumPy) raising dataset reliability above 98%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Video Analytics | Hyderabad, India

- Built transformer-based video summarization system; reduced manual review time 60-70% and achieved ~85% alignment with human-curated highlights.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30% and raised contextual accuracy >85%.
- Deployed Flask-based API with lightweight web interface for non-technical staff adoption.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP, Industrial Automation | Hyderabad, India

- Optimized SQL/T-SQL stored procedures; reduced query execution time ~20% and latency ~25%.
- Built CI/CD pipelines with Jenkins; reduced deployment errors >30%, improved release cycle time ~35-40%.
- Designed Grafana dashboards tracking long-running queries and deadlocks; cut incident recurrence ~25%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization to Fraud Risk Scoring) with schema-validated JSON contracts, RAG-grounded validation, and audit-ready reasoning traces. Strong POC-to-production pattern applicable to Boomi customer deployments.
Python, LangChain, LlamaIndex, FastAPI, Docker, RAG

Dream Decoder - Multi-API AI Orchestration Platform

Full-stack FastAPI + React/TypeScript app coordinating multiple AI APIs (Perplexity Sonar, Replicate image models); intermediate structured prompt transformation layers improved contextual alignment ~**30%** and first-pass success rate ~**25-30%**. Demonstrates multi-vendor AI integration -- directly relevant to Boomi's integration-platform context.
FastAPI, React, TypeScript, Vite, Perplexity Sonar API, Replicate

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Production browser extension integrating 4 AI vision providers; built and shipped independently with 29-site support. High-agency product delivery example.
TypeScript, Claude Sonnet, GPT-4o mini, multimodal AI

EDUCATION

Master's Degree - *Kent State University, Ohio, USA*

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, guardrails

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, embeddings, vector search

Infra & Integration: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Enterprise AI Automation, iPaaS/Integration AI, Healthcare AI, Workflow Automation